

EDUCATION	PhD in Evolutionary Genomics 2017 - 2022 Centre for Biological Diversity, University of St Andrews , Scotland Research project: <i>Short-term adaptation in the lab: an analysis of evolutionary trajectories using time-series data</i> Supervised by Dr. Carolin Kosiol
	MSc in Evolutionary and Developmental Biology 2014 - 2017 Faculdade de Ciências da Universidade de Lisboa , Lisbon, Portugal Research project: <i>Comparative genomic analyses of cyanobacteria</i> Supervised by Professor Octávio Paulo (CoBiG2, CE3C) and Dr. Elsa Dias (INSA) Average: 18/20
	MSc in Evolutionary Genetics 2015 - 2016 Institute of Evolutionary Biology, University of Edinburgh , Scotland Research project: <i>Searching for selective sweeps in the genome of the wild house mouse, <i>Mus musculus castaneus</i></i> Supervised by Professor Peter Keightley (IEB) and Dr. Tom Booker with Distinction
	BSc in Biology - Evolutionary and Developmental Biology 2011 - 2014 Faculdade de Ciências da Universidade de Lisboa , Lisbon, Portugal Average: 16/20
PUBLICATIONS	
Journal publications	Barata, C , Borges, R, and Kosiol, C. – Bait-ER: a Bayesian method to detect targets of selection in Evolve-and-Resequencing experiments. (<i>in review</i>) bioRxiv: doi.org/10.1101/2020.12.15.422880
	Barata, C , Snook, R, Ritchie, M, Kosiol, C. – Selection on the fly: short term adaptation to an altered sexual selection regime in <i>Drosophila pseudoobscura</i> (<i>in review</i>) bioRxiv: doi.org/10.1101/2022.03.16.484639
Conferences and meetings	55th Population Genetics Group Meeting – PopGroup55 Norwich 2022, Online meeting January 2022 Contributed talk – <i>Selection on the fly: genomic time series analysis of adaptation in <u>D. pseudoobscura</u></i>
	SMBEv2021 Meeting Society for Molecular Biology & Evolution meeting, Online meeting July 2021 Contributed talk – <i>Predicting the effects of selection in the lab: a sexual selection experiment in <u>D. pseudoobscura</u></i>
	EMBO Workshop: Predicting Evolution European Molecular Biology Organisation workshop, Online meeting June 2021 Poster presentation – <i>Predicting the effects of altered sexual selection on the genome: An experiment in <u>Drosophila pseudoobscura</u></i>
	54th Population Genetics Group Meeting – PopGroup54 Liverpool 2021, Online meeting January 2021 Contributed talk – <i>Bait-ER: fishing for selection in a pool of drifting alleles</i>

25th DZG Graduate Meeting

Online meeting | September 2020

Contributed talk – *Bait-ER: fishing for selection in a sea of alleles*

A Century of Genetics

Royal College of Physicians, Edinburgh | November 2019

Poster presentation and Poster pitch – *Bait-ER: fishing for targets of selection in a sea of alleles*

5th Probabilistic Modeling in Genomics Conference (ProbGen19)

Aussois, France | October 2019

Poster presentation – *Bait-ER: fishing for targets of selection in a sea of alleles*

Evolution 2019

Rhode Island Convention Centre, Providence, RI, USA | June 2019

Invited student speaker, Spotlight Session – *Modelling N_e to infer evolutionary trajectories from time series data*

Workshop on Mathematical and Statistical Aspects of Molecular Biology (MASAMB) - 30 years edition

EMBL-EBI, Wellcome Genome Campus, Cambridge | April 2019

Contributed talk – *Estimating selection coefficients in Experimental Evolution*

52nd Population Genetics Group Meeting – PopGroup52

Oxford Brookes University, Oxford | January 2019

Poster presentation

StABU Bioinformatics Workshop

St Andrews Bioinformatics Unit, University of St Andrews | September 2018

Oral presentation

II Joint Congress on Evolutionary Biology - Evolution 2018

Montpellier, France | August 2018

Poster presentation – *Modelling N_e to infer evolutionary trajectories from genome-wide data*

28th Annual Workshop on Mathematical and Statistical Aspects of Molecular Biology (MASAMB)

University of St Andrews | March 2018

Poster presentation, Organising Committee

GRANTS AND AWARDS

Sir Ken Murray Endowment Fund (2022): PhD Student Conference Grant | £500 to cover registration fees, accommodation and travel expenses.

Genetics Society Conference Grant (2019): Junior Scientist Conference Grant, non-GS Meeting | £740 to cover registration fees, accommodation and travel expenses.

ESEB Ad-Hoc Funding and Genetics Society One-Off Meeting Sponsorship Grant (2019): for organising a workshop at the University of St Andrews with Alberto Carmagnini (Queen Mary University of London), Bernardo Gutierrez (University of Oxford) and Jessica King (University of Edinburgh) - 3000€ from ESEB and £1900 from the Genetics Society.

Evolution 2019 Travel Award: Travel Stipend from SSE (Society for the Study of Evolution) | \$500 to cover travel expenses.

A Century of Genetics – Poster Prize (2019): Prize awarded for second best poster at "The Germline, Sex Determination & Sex Chromosomes" session | £180 monetary prize.

COMPUTER SKILLS

Programming languages: Bash, Python, R

Bioinformatics tools: forward-time and coalescent simulations, genome-wide selection scans, phylogenetic inference, genome assembly and mapping, variant calling

Operating systems: UNIX, Windows

Miscellaneous: Experience with the use of High Performance Clusters (Sun Grid Engine and Slurm Workload Manager)

TEACHING

Demonstrating & tutoring: Biology 2, Computational Genomics, Ecology, Evolution, and Evolutionary Biology (over 350 hours) | University of St Andrews

ACADEMIC SERVICE

BioNet Committee Member (February 2018 - December 2021): Postgraduate Biology Network (BioNet) President, Secretary and Treasurer | University of St Andrews

Genetics Journal Club (February 2019 - January 2020): organised a monthly evolutionary genetics journal club | Centre for Biological Diversity, University of St Andrews

Fantastic Forces workshop (June 5th-7th 2019): Organised a 3-day student-led workshop on new-developments in Evolutionary Genetics for postgraduate students at the University of St Andrews in collaboration with other UK universities. Prepared an afternoon tutorial session on how to estimate selection coefficients using genome-wide laboratory time series data.

PhD Class Representative (September 2019 - October 2021): Class rep for PhD students in the School of Biology | University of St Andrews

Paper reviewer for relevant journals: Molecular Ecology Resources